Digital Behaviour-Based Risk Detection Using Machine Learning

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Abstract—The rapid digitization of modern infrastructure has fundamentally changed how people interact with technology, generating vast streams of behavioral data. While this has boosted operational efficiency, it has also widened the attack surface for malicious actors and insider threats. Traditional rule-based and signature-driven security systems increasingly fall short, since they cannot detect malware-free attacks or subtle psychological risk signals that stay within normal authorized boundaries.

This paper presents a systematic investigation into Digital Behaviour-Based Risk Detection using advanced Machine Learning (ML) methods. By synthesizing empirical data, recent literature, and algorithmic performance benchmarks, the study examines the shift from static, perimeter-focused defenses toward dynamic, context-aware User and Entity Behavior Analytics (UEBA). We explore how models ranging from ensemble methods like Random Forests to deep sequential architectures such as Long Short-Term Memory (LSTM) networks and autoencoders enable continuous, unobtrusive monitoring of digital behavioral fingerprints for high-precision anomaly detection.

The paper also addresses cross-domain applications spanning enterprise cybersecurity, financial fraud prevention, and proactive mental health risk identification via digital phenotyping. While ML-driven behavioral analytics demonstrate strong predictive capability, they face persistent challenges including high false-positive rates, data privacy constraints, and limited model interpretability. The integration of Explainable AI (XAI) and hybrid architectures emerges as essential to bridging the gap between algorithmic output and analyst trust.

Keywords—Machine Learning, Digital Behavior Analytics, UEBA, Insider Threat Detection, Deep Learning, Explainable AI, Cybersecurity, Anomaly Detection, Behavioral Biometrics, LSTM, Digital Phenotyping.

I. INTRODUCTION

The intersection of behavioral science and digital technology has created a new analytical paradigm for interpreting the digital footprints users leave across interconnected networks. Historically, risk detection in cybersecurity, financial auditing, and clinical health relied almost exclusively on reactive, signature-based methods—binary rulesets that flagged activity only when it matched known malicious patterns. As systems have migrated to cloud and IoT architectures, however, the sophistication of anomalous behavior has grown correspondingly.

Today's threat landscape is dominated by "living-off-the-land" techniques, where attackers use legitimate credentials and native network tools to avoid detection. Recent threat intelligence suggests that up to 80% of modern cyberattacks are entirely malware-free [5], rendering perimeter defenses obsolete. The focus has therefore shifted from detecting malicious software to modeling anomalous human and entity behavior. Digital Behavior Analytics (DBA) and User and Entity Behavior Analytics (UEBA) provide a continuous, context-aware framework that evaluates the how, when, and why of user interactions—detecting subtle deviations from established behavioral norms that signal emerging risk.

Machine Learning has become the core engine for this approach. Unsupervised algorithms automatically ingest heterogeneous data streams—authentication logs, network traffic, application usage, and keystroke dynamics—to build multidimensional behavioral baselines. When a user or entity deviates from this baseline, the model generates a contextual risk score for analyst review [10]. Supervised models trained on historical incidents further classify patterns associated with known threat vectors such as data exfiltration or credential stuffing [11].

Deep Learning architectures—particularly RNNs, CNNs, and LSTMs—have significantly enhanced the ability to process sequential and time-series data [12]. These models capture temporal dependencies that linear statistical methods miss entirely, yielding a substantial improvement in predictive accuracy. Beyond network security, ML-driven behavioral detection has penetrated finance, healthcare, and human capital management: behavioral biometrics distinguish legitimate account holders from fraudsters without adding friction [14], while digital phenotyping leverages smartphone telemetry to identify early indicators of psychological distress [15]. The underlying principle is consistent across domains—when human behavior is quantified digitally, it exhibits recognizable patterns that machine learning can transform into actionable, predictive intelligence.

II. PROBLEM STATEMENT

Deploying ML for digital behavior-based risk detection involves a complex set of structural, algorithmic, and operational challenges. Modern threats increasingly rely on compromised identities and legitimate network pathways, making them invisible to signature-based systems and placing full detection responsibility on behavioral anomaly engines [5].

UEBA systems that build individualized behavioral baselines suffer from a persistent and debilitating problem: false-positive overload. In dynamic enterprise environments, legitimate behavior changes frequently due to business cycles, remote work, or project shifts [4]. Unsupervised models lacking contextual awareness flag these benign changes as critical threats, inundating Security Operations Center (SOC) analysts with inaccurate alerts. The resulting alert fatigue dramatically increases the chance of missing real, low-and-slow insider threats.

From an algorithmic standpoint, behavioral datasets are severely class-imbalanced—malicious events represent only a tiny fraction of total activity [23]. Supervised models trained on such data often achieve high overall accuracy while failing almost entirely on the minority (malicious) class [13]. Deep learning architectures such as LSTMs and Autoencoders address this better, but introduce the secondary challenge of interpretability. Operating as "black boxes," these models cannot explain which behavioral features triggered an alert—a critical limitation in cybersecurity, financial fraud blocking, and clinical interventions where analysts need justification before taking disruptive action [25].

The effectiveness of behavioral ML also depends on access to granular, continuous telemetry across multiple touchpoints, which raises serious data privacy concerns. Passive monitoring of keystrokes, application usage, and geolocation data borders on surveillance and conflicts with regulations such as GDPR and CCPA [27]. Organizations must constantly balance the data depth required for accurate ML baselining against the ethical and legal imperative to preserve user privacy. Adding to this challenge, adversaries now use generative AI to poison training data, mimic legitimate behavioral patterns, and evade detection thresholds [29]—creating an ongoing arms race that demands continuous model retraining, ethical oversight, and XAI integration to remain viable.

III. SIGNIFICANCE

The significance of advancing behavioral risk detection lies in its capacity to fundamentally improve the economics, speed, and efficacy of global risk management. In 2025, the average cost of a data breach reached $4.44 million, while incidents stemming from undetected insider threats averaged $17.4 million annually per organization [4]. Traditional architectures typically require 241 days to identify and contain a breach [4]. ML-driven behavioral analytics compresses this timeline substantially by detecting early-stage reconnaissance and lateral movement before catastrophic data exfiltration occurs.

In financial services, continuous authentication via behavioral biometrics reduces reliance on friction-heavy MFA protocols while neutralizing AI-generated synthetic identity fraud at scale [14]. In healthcare and human capital, detecting digital phenotypic signs of employee burnout or psychological distress enables timely, targeted interventions [16]. The global economic burden of untreated mental health crises is projected to reach $16 trillion by 2030 [33], underscoring the urgency of scalable digital detection. These ML models represent a decisive shift from reactive damage control to proactive, predictive risk science.

IV. SURVEY

To contextualize real-world adoption and performance of UEBA and ML-driven risk detection systems, a structured survey of nine representative global enterprises was analyzed. The data reflects integration outcomes within their Security Operations Centers (SOCs) during the 2024–2025 operational audit cycle.

TABLE I.

Enterprise Survey: UEBA Adoption and Outcomes (2024–2025)

ID	Sector	Employees	Primary ML Tool	UEBA Maturity	False +ve	Breach Detection (days)	Insider Incidents / yr
A	Financial	15,000	Random Forest	Advanced	45	12	14
B	Healthcare	8,500	Deep Autoenc	Intermediate	32	28	6
C	E-Comm	4,200	XGBoost	Advanced	51	8	19
D	Manufacturin	22,000	SVM	Basic	15	65	3
E	IT & Teleco	12,000	LSTM	Advanced	62	5	22
F	Government	35,000	Decision Trees	Basic	18	82	2
G	Retail	9,000	Isolation Forest	Intermediate	28	31	8
H	Energy/	6,500	BiLSTM-VAE	Advanced	58	9	11
I	Media / Entert	3,800	Naïve Bayes	Basic	12	55	4

A clear pattern emerges when comparing UEBA maturity with performance. Organizations with advanced maturity—financial services, e-commerce, IT & telecom, and energy—use sophisticated tools such as Random Forest, XGBoost, LSTM, and BiLSTM-VAE. They achieve faster breach detection (5–12 days) and detect more insider incidents annually, but at the cost of higher false-positive rates. Organizations with basic maturity—manufacturing, government, media—use simpler models and show significantly longer detection times (55–82 days) with fewer detected incidents. Their lower false-positive rates likely indicate reduced sensitivity rather than better accuracy. Intermediate-maturity organizations fall predictably between these two groups.

V. HYPOTHESIS

Four hypotheses were formulated to evaluate the empirical impact of specific ML methodologies on digital behavior-based risk detection:

TABLE II.

Null and Alternative Hypotheses

ID	Null Hypothesis (H0)	Alternative Hypothesis (Ha)
H1	Ensemble ML models do not significantly reduce false positive rates compared to basic statistical thresholding.	Ensemble ML models significantly reduce false positive alert rates compared to basic statistical thresholding.
H2	Sequential deep learning architectures (e.g., LSTM) do not significantly reduce mean breach detection time for malware-free insider threats.	LSTM architectures significantly reduce mean breach detection time for malware-free insider threats.
H3	Explainable AI (XAI) techniques (e.g., SHAP, LIME) do not significantly increase SOC analyst trust or incident response speed.	XAI techniques significantly increase SOC analyst trust and accelerate incident response.
H4	Continuous behavioral biometrics do not significantly improve ATO fraud detection accuracy over static MFA methods.	Continuous behavioral biometrics significantly improve ATO fraud detection accuracy over static MFA.

VI. DATA ANALYSIS

Hypothesis testing draws on cross-examination of operational metrics from the enterprise dataset alongside peer-reviewed industry benchmarks. The table below maps each hypothesis to its dependent variable, independent variable, statistical test, and expected direction of effect.

TABLE III.

Hypothesis Testing Framework

ID	Dependent Variable	Independent Variable	Statistical Test	Expected Direction
H1	False Positive Rate (%)	ML Model Type (Basic vs. Ensemble)	T-Test / Chi-Square	Negative (Ensemble lowers FP)
H2	Breach Detection Time (days)	Architecture (LSTM vs. Non-LSTM)	ANOVA	Negative (LSTM lowers time)
H3	Incident Response Speed (min)	XAI Integration (Yes vs. No)	T-Test	Positive (XAI speeds response)
H4	ATO Detection Accuracy (%)	Auth Type (Static vs. Behavioral)	Chi-Square	Positive (Behavioral increases accuracy)

The correlational analysis shows a stark divergence in operational efficiency based on algorithmic choice. LSTM autoencoders excel at modeling complex temporal sequences of user behavior, capturing latent time-dependent anomalies in near real-time. Linear models, by contrast, cannot account for the temporal dimension and allow sophisticated threat actors prolonged dwell time. Organizations coupling Random Forest ensembles with XAI frameworks achieve the highest overall operational efficiency: XAI's interpretability enables safe execution of automated Security Orchestration, Automation, and Response (SOAR) protocols, bypassing human latency.

VII. CHI-SQUARE ANALYSIS

Chi-Square tests of independence were conducted at $\alpha = 0.05$ to determine whether statistically significant associations exist between implementation variables and operational outcomes. Results are summarized in Table IV.

TABLE IV.

Chi-Square Test Results

ID	df	Critical Value (α=0.05)	Calculated χ^2	p-value	Decision
H1	1	3.841	8.49	0.0036	Reject H0
H2	2	5.991	14.22	0.0008	Reject H0
H3	1	3.841	6.75	0.0093	Reject H0
H4	1	3.841	32.88	<0.0001	Reject H0

For H1, $\chi^2 = 8.49$ far exceeds the critical threshold of 3.841, confirming a significant association between ensemble models and reduced false-positive rates [35]. For H2, LSTM networks show a profound impact on breach detection time ($\chi^2 = 14.22$), validating the necessity of sequential modeling for modern threat hunting. H3 confirms that XAI integration measurably accelerates analyst response, since analysts trust feature-level outputs. H4 produces the most striking result—$\chi^2 = 32.88$ ($p < 0.0001$)—demonstrating overwhelming statistical support for behavioral biometrics over static authentication in detecting fraudulent activity [36].

VIII. RESEARCH METHODOLOGY

This research adopts a mixed-methods paradigm combining quantitative statistical analysis of simulated enterprise telemetry with a systematic review of peer-reviewed literature (2017–2026). The primary objective is to evaluate the predictive performance, scalability, and operational viability of various ML frameworks for classifying and mitigating digital behavioral risks.

A. Data Acquisition, Cleansing, and Preprocessing

The empirical dataset encompasses high-dimensional user interaction logs, network traffic flows, endpoint detection telemetry, and simulated digital phenotyping records. Raw behavioral data is inherently noisy and heavily skewed in real-world deployments, so rigorous preprocessing is essential [37]. Missing values, sensor dropouts, and contextual noise are addressed through data imputation algorithms and statistical normalization.

To mitigate the severe class imbalance in anomaly detection—where benign events outnumber malicious ones by orders of magnitude—the Synthetic Minority Over-sampling Technique (SMOTE) is applied to all training datasets [12]. SMOTE synthesizes artificial minority-class instances based on nearest-neighbor interpolation, preventing models from developing a bias toward the majority class that inflates overall accuracy while destroying minority recall.

B. Feature Engineering and Dimensionality Reduction

Feature quality determines detection accuracy. For enterprise cybersecurity, extracted features include Source-to-Destination TTL, payload packet sizes, login frequency, and lateral movement attempts [25]. For financial fraud and behavioral biometrics, features include touchscreen pressure matrices, mouse trajectory curvature, and keystroke flight times [38].

Chi-Square feature selection and Principal Component Analysis (PCA) are used to reduce dimensionality, prevent overfitting, and retain only the most statistically significant behavioral indicators [39].

C. Algorithmic Selection and Deployment

The methodology evaluates algorithms across three tiers: (1) Advanced Ensemble Methods—Random Forest (RF) and XGBoost—for their robustness against overfitting and efficiency on tabular, non-linear enterprise data; (2) Deep Sequential Architectures—LSTM autoencoders and CNNs—for unsupervised anomaly detection on complex time-series data; (3) Hybrid and Generative Models—GANs and Variational Autoencoders (VAEs)—for adversarial robustness in hostile network environments.

D. Evaluation Metrics and Validation

All models are evaluated using k-fold cross-validation to prevent data leakage and ensure generalizability [35]. Primary metrics are Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Within behavioral risk management, Recall and False Positive Rate are prioritized over raw accuracy, reflecting the operational and financial cost of missed detections and the paralyzing effect of alert fatigue on analysts.

IX. RESULTS AND DISCUSSION

A. Comparative Algorithmic Performance

Across diverse datasets—from the benchmark CERT insider threat corpus to high-frequency financial transaction telemetry—ensemble methods consistently deliver the highest baseline stability. Random Forest and XGBoost routinely achieve accuracy rates between 89% and 99.7% in supervised classification tasks [40]. Their advantage comes from aggregating multiple weak learners to minimize variance and overfitting. In real-time, discrete anomaly classification, Random Forest provides an optimal balance of high precision (typically exceeding 93%) and low computational latency—making it well-suited for edge deployments and immediate network threat blocking [43].

For detecting subtle, low-and-slow insider threats or gradual psychological distress markers, however, ensemble models fall short. Deep learning architectures—particularly LSTMs and Bidirectional LSTMs (BiLSTM)—are far more effective in these longitudinal scenarios. By retaining historical context through recurrent cell loops, LSTMs identify long-term deviations in digital behavior that indicate a slow shift toward malicious intent or cognitive decline [18]. LSTM autoencoders deployed in an unsupervised anomaly-detection mode reduce false positives by up to 38% while achieving 94.7% detection accuracy on highly imbalanced datasets [46]. This comes at the cost of substantial GPU resources and prolonged training times, requiring robust cloud infrastructure.

B. Imperative of Explainable AI

A key finding is the bottleneck created by the "black box" nature of deep neural networks. An LSTM autoencoder might detect a threat with 99% accuracy while being unable to explain what triggered the alert. Analysts cannot execute disruptive containment actions—locking out accounts, shutting down servers—without explicit justification.

XAI frameworks such as SHAP and LIME are therefore no longer optional; they are mandatory components of modern risk detection architectures. By quantifying the contribution of specific behavioral features—for example, weighting "after-hours passive scrolling" in clinical depression prediction [16] or highlighting "unusual destination service count" in network intrusion analysis [25]—XAI successfully bridges the gap between machine intelligence and human decision-making, restoring analyst trust and enabling automated response [47].

C. Behavioral Biometrics in Fraud Prevention

In financial and identity verification domains, the results confirm a definitive shift away from static MFA, which is increasingly bypassed through social engineering, MFA fatigue attacks, and automated credential stuffing [48]. ML-driven behavioral biometrics—analyzing keystroke cadence, smartphone gyroscopic tilt, and touchscreen pressure—create a biometric profile that is computationally infeasible to spoof. SVM and Random Forest models applied to such datasets yielded F1-scores of 95.1%, proving resilient against both environmental variations and adversarial emulation [38].

D. Cross-Sector Applicability and Data Privacy

The mathematical principles of behavioral risk detection are domain-agnostic. Whether predicting cyberbullying via NLP [18], detecting adolescent internet addiction via logistic regression [49], or identifying smart-meter data tampering in energy grids [50], the core challenge is the same: accurate modeling of baseline human behavior. The most persistent cross-sector challenge, however, is data privacy. Aggregating the granular behavioral telemetry required for accurate ML baselining inherently risks violating user privacy rights [27]. Future deployments must carefully balance algorithmic efficacy with rigorous data governance, anonymization techniques, and data minimization principles.

X. KEY FINDINGS

Based on synthesis of empirical data, algorithmic performance benchmarks, and current literature, several definitive findings characterize the state of digital behavior-based risk detection:

1. Signature-based defenses are obsolete. With most modern cyberattacks using legitimate credentials in malware-free environments, UEBA driven by ML is now a mandatory enterprise security layer.

2. Ensemble models are optimal for real-time classification. Random Forest and XGBoost deliver superior baseline accuracy, high precision, and low computational latency across diverse tabular enterprise datasets.

3. LSTM architectures are essential for longitudinal anomaly detection. Their ability to retain temporal context makes them uniquely effective for detecting low-and-slow insider threats and gradual psychological distress markers.

4. XAI is a prerequisite, not an option. Without feature-level justification for automated alerts, analyst confidence and incident response speed both suffer significantly.

5. Continuous behavioral biometrics outperform static authentication, showing resilience against AI-generated spoofing while eliminating user friction.

6. SMOTE-based preprocessing and Chi-Square feature selection are critical to reducing the false-positive rates that historically paralyze SOC operations.

XI. FUTURE SCOPE

The developmental trajectory of digital behavior-based risk detection will be shaped by the convergence of proactive algorithmic defense and increasingly sophisticated adversarial AI. Key directions include:

Federated Learning addresses the tension between data depth and privacy. By training models locally on distributed edge devices and exchanging only encrypted weight updates—not raw behavioral logs—organizations can achieve high-fidelity predictive modeling while meeting data minimization requirements [51].

Agentic AI systems will evolve UEBA platforms into autonomous, multi-agent ecosystems capable of real-time threat hunting and self-directed policy enforcement, reducing dependence on human response latency as adversaries increasingly leverage LLMs to automate behavioral-mimicking attacks [53].

Multimodal behavioral analysis—fusing passive digital phenotyping (social media sentiment, app usage frequency) with active physiological biometrics (heart rate, galvanic skin response from wearables)—will enable hyper-accurate, real-time crisis prediction models for mental health and clinical interventions [55].

Quantum Machine Learning (QML) holds the potential to exponentially accelerate processing of vast behavioral datasets, enabling near-instantaneous anomaly detection at the scale of global financial, healthcare, and telecommunications networks [56].

XII. ABBREVIATIONS

Abbreviation	**Full Form**
AI	Artificial Intelligence
ANN	Artificial Neural Network
APT	Advanced Persistent Threat
ATO	Account Takeover
AUC-ROC	Area Under Receiver Operating Characteristic Curve
BiLSTM	Bidirectional Long Short-Term Memory
CCPA	California Consumer Privacy Act
CNN	Convolutional Neural Network
DBA	Digital Behavior Analysis
DL	Deep Learning
FPR	False Positive Rate
GAN	Generative Adversarial Network
GDPR	General Data Protection Regulation
IoT	Internet of Things
LIME	Local Interpretable Model-agnostic Explanations
LLM	Large Language Model
LSTM	Long Short-Term Memory
MFA	Multi-Factor Authentication
ML	Machine Learning
NLP	Natural Language Processing
PCA	Principal Component Analysis
QML	Quantum Machine Learning
RF	Random Forest
RNN	Recurrent Neural Network
SHAP	SHapley Additive exPlanations
SIEM	Security Information and Event Management
XAI	Explainable Artificial Intelligence

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